TEACHER GUIDE / BREADBOARD

Teaching with this workbook

Edition 1.0. The student PDF is planned as 50 A4 pages for 25 double-sided sheets. The editable Word edition must be checked in native Word before release because pagination can change. This guide supplies answers and expected evidence; it is not a new assessment schedule or machine operating procedure.

Use the source course as the lesson hub

All fifteen named theory sections are taught in three reading pages per module. The complete thirty-question source MCQ set is retained in original order with original options. All eleven source extended responses are retained: Module 5 contains an additional peer-feedback task. The twelve original folio checkpoints have paper planning/evidence spaces and an index.

Keep demonstrations and practical authority clear

Use the five course slide decks and the six selected course video clips with a specific observation focus. The original one-page supplied Breadboard plan, current teacher directions, school procedures and approved product information control practical work. Do not treat a photo, conceptual diagram or a workbook answer as permission for machine use.

Before delivery

Confirm current plan interpretation, stock, material, thickness, joint quantity and locations, adhesive, finish, machine setup, supervision, risk controls, school first-aid process, storage and waste arrangements. Confirm assessment marks, dates, conditions and submission requirements separately. None is created by this workbook.

Photographic evidence

The course folio asks for photographs of students' own work. The paper spaces permit prints, notes and labelled sketches to support a caption; they do not remove the photograph requirement. Agree how images are captured and referenced. Accept honest missing/pending status and arrange teacher guidance; never ask students to simulate a past check as original evidence.

TEACHER GUIDE / BREADBOARD

Sources and illustration limits

Source snapshot: https://stevencowell.github.io/bread-board-guided-course/ at repository commit 21df8330f6bb5b11771a4350771cbcdf1df55ad9. Primary teaching: modules/module-01.html through module-05.html, theory-1 to theory-3. Question and model-answer source: matching module-XX-data.js files. Folio: breadboard-folio.html. Presentation source: presentations/breadboard-module-01.pptx through 05. Teacher planning context: the v2.1 alignment manifest and program.

The current HTML includes first aid/WHS material and the current Module 5 questions include peer feedback. Older markdown and the incomplete source peer-feedback model do not override these explicit current requirements. The supplied plan is assets/breadboard-plans.pdf, titled Breadboard, WWHS, November 2012; reproduced unchanged on student page 3.

The source hero is an illustrative completed project image. A new generated grain study shows complete unassembled parts and grain contrast. Bespoke labelled vector diagrams explain datum, matching, preparation, checking, resource care and evidence relationships. They are not approved construction plans, machine setups or student work. The commercial engraved reference image is excluded because its handle, decoration and form do not match the supplied project. The source finishing-waste image is excluded because its open bin could imply an unsupported disposal method.

No timber species, thickness, biscuit size, slot depth, machine setting, abrasive grit sequence, adhesive time, oil product, coat count, food-safety claim or formal assessment marks have been invented. Where required values are unavailable, student records ask for teacher confirmation.

Use Australian English. For open tasks, judge the reasoning, observable evidence and relevant source relationship. Do not require the wording of an example answer. Discuss misconceptions and permit correction after students return to the reading. Practical sign-off is separate from completing a worksheet.

TEACHER GUIDE / BREADBOARD

Pages 7–8: answers and evidence

Student page 7: Check readiness

1. B. The brief, plan and teacher directions define this project.

2. A. Correct. A hazard is a source or situation with the potential to cause harm.

3. C. Correct. Stop, protect others and report the fault.

Use the teacher-approved safe response: remove/resolve the obstruction when safe and directed; stop use, keep others clear and report the damaged tool; stop work, alert the teacher and follow school first-aid arrangements. Do not invent a repair or treatment.

Student page 8: Check your marks

4. B. Correct. One reliable reference reduces cumulative error.

5. B. Correct. One fine checked line gives a clear boundary.

6. C. Correct. Clarify before continuing.

Order in the printed list: 4, 1, 5, 2, 3. Read the plan; establish datum; measure; recheck; then make/check the clear mark before any removal.

The final comparison/recheck before removal establishes clear correct references and a fine intended boundary. Pencil errors remain reversible; removed timber cannot simply be restored.

TEACHER GUIDE / BREADBOARD

Pages 9–10: answers and evidence

Student page 9: Explain your decisions

Expected content: brief or plan, personal readiness, workspace hazards, report or stop, first aid response. I will first read the project brief, plan and teacher directions so I understand the required stage. I will secure hair and loose items, wear the required PPE and check that my clothing and footwear are suitable. I will inspect the bench, floor and nearby walkway for clutter, spills, damaged equipment or loose material. If I find damage or cannot make a condition safe, I will stop, keep others clear and report it to the teacher. If an injury or sudden illness occurs, I will stop work, alert the teacher immediately, keep the area safe and obtain the trained first aider through the school procedure. I will not give treatment beyond my training or include private medical details in my work.

Expected content: read the plan, datum, measure and check, fine line, stop before cutting. I would read the written plan information and identify exactly where the measurement begins and ends. I would use the agreed datum or reference edge for every related measurement so the positions remain aligned. I would measure from that datum twice, view the scale carefully and make one fine line with a sharp pencil. I would check that the line is square or correctly positioned and compare it with the plan. If anything is unclear, I would stop and ask the teacher before removing material.

Student page 10: Brief and safety evidence

Student identifies real brief/plan requirements, not invented choices; proposed checks relate specifically to those requirements. Missing information is marked teacher to confirm.

Drawing is a conceptual own-work layout, showing a realistic hazard and suitable teacher-approved control; avoid depicting unsafe machine use. Explain clear access and current materials only.

Specific actual readiness/shared-space check and reason/result. Plan/brief evidence supports first decision; real photo reference or honest pending status; no private incident details.

TEACHER GUIDE / BREADBOARD

Pages 11–16: answers and evidence

Student page 11: A datum you can explain

A clear datum and consistent related position; reference face/edge and component orientation identified. Drawing is explanatory, not a substitute construction plan. Do not require invented sizes.

Actual plan/layout and datum/fine-line checks; record correction if made, otherwise actual result. Uncertainty goes to teacher before removal.

Specific checked datum-based mark, how ambiguity was prevented, how result agrees with approved source; actual photo reference or honest pending status.

Student page 15: Check the layout

7. B. Correct. Grain affects appearance and how timber behaves.

8. B. Correct. Reference marks and pairing preserve orientation.

9. B. Correct. Transferring marks keeps the slots aligned.

Re-establish intended orientation against plan; identify matching reference faces/edges, paired components and transferred centre lines; do not rely on visual similarity. No machining until correspondence is checked and uncertainty resolved.

Student page 16: Check controlled shaping

10. B. Correct. Follow the plan and teacher direction.

11. A. Correct. The visible line protects the intended form.

12. A. Correct. Stop and seek direction when control or clarity is lost.

Correction should respond to the observed fault, not force both parts through identical removal. Preserve the sound component and planned boundary; check the cause/high area with teacher, use approved controlled correction if authorised and compare again.

TEACHER GUIDE / BREADBOARD

Pages 17–19: answers and evidence

Student page 17: Explain your decisions

Expected content: grain inspection, plan orientation, reference marks, paired components, final check. I will inspect the timber for grain direction, visible defects and appearance before arranging the components. I will use the project plan to keep the central board and curved end pieces in the required order and orientation. I will label the reference faces or edges and keep matching parts paired while marking. Before any material is removed, I will compare the layout with the plan, check that the intended faces remain clear and ask the teacher if a defect or orientation is uncertain.

Expected content: plan and teacher, aligned components, transferred centre lines, reference orientation, check before machining. The approved biscuit-joint positions come from the project plan and teacher directions, not from guesswork. I will place the matching components in their correct orientation and align their reference faces and edges. I will transfer each centre line across the meeting parts while they are held together, then label the marks so the matching slots cannot be confused. Before any slot is made, I will check the number, location, spacing and orientation against the plan and stop for teacher advice if a mark is unclear.

Student page 18: Select and arrange the timber

Specific own-stock observations, linked to orientation/appearance/working behaviour, intended role and plan. Defects are identified and teacher advice sought; do not claim species or suitability from the illustration alone.

Reasoned grain/defect/appearance/component-purpose selection from own stock with accurate teacher advice or pending status. Preserve approved plan and reference orientation.

Student page 19: Plan the matching relationship

Matching component faces/edges and shared centre line align; biscuit is flat oval, not dowel. Diagram has no invented number/spacing/depth. A complete practical setup must be teacher-confirmed.

Own authorised shaping/preparation stage, visible planned boundary/quality check, reason and actual result; no unsafe invented operation.

Correct component pair, orientation, reference faces/edges and aligned transferred centre lines, compared with plan before slots. Photo is own real work or marked pending.

TEACHER GUIDE / BREADBOARD

Pages 23–24: answers and evidence

Student page 23: Check before joining

13. B. Correct. It checks the complete project while correction is still possible.

14. C. Correct. Diagnose before changing the component.

15. A. Correct. Planning reduces rushing and missed checks.

Inspect correct orientation, matching marks, debris/contact and local high points before correction. For rocking inspect full assembly alignment/support and possible twist; do not assume its cause or apply force. Teacher advice where uncertain.

Student page 24: Check the preparation

16. B. Correct. Pressure should close verified joints while preserving alignment.

17. A. Correct. Authorisation, support and clear preparation come first.

18. B. Correct. Stop and seek authorised direction.

Stop; move away safely as directed; ask teacher to review support, component orientation and intended profile; restart only after the setup and method are clarified and authorised. Slowing down does not fix instability.

TEACHER GUIDE / BREADBOARD

Pages 25–26: answers and evidence

Student page 25: Explain your decisions

Expected content: compare with plan, inspect fit, do not force, controlled correction, teacher advice. I would stop and remove the assembly without forcing the end piece. I would confirm its orientation against the plan and inspect the meeting surfaces for a gap, high spot, debris or a mark that does not align. I would compare both ends and use the dry fit to identify where contact occurs. Only after the cause was verified would I remove a very small amount using the approved method and test the assembly again. If the cause remained unclear or the component appeared damaged, I would ask the teacher before making another change.

Expected content: dry fit passed, parts and clamps ready, approved adhesive, balanced pressure, alignment and clean-up. Before adhesive is applied, I will confirm that the complete dry assembly fits, aligns with the plan and has the correct orientation. I will arrange the parts, clamps, protective pieces and checking tools in the intended sequence. I will use only the teacher-approved adhesive and follow the demonstration and label directions. I will apply controlled, balanced pressure so the joints close without pulling the board out of alignment. I will recheck the overall form as pressure is added and manage excess adhesive using the approved method before leaving the assembly as directed.

Student page 26: Record the complete dry fit

Authentic complete dry fit shows required relationships. Label checks in own assembly; print or drawing plus source-required real photograph reference. No invented passed result.

Actual observed gaps/contact/orientation/alignment/stability; specific appropriate action and recheck, teacher if unexplained. Accept no fault observed with evidence; reject forced fit or random material removal.

Dry fit preserves opportunity to separate, diagnose and correct before permanent joining; whole assembly confirmation, not a single joint only.

TEACHER GUIDE / BREADBOARD

Pages 27–32: answers and evidence

Student page 27: Record the glue-up decisions

Actual successful dry-fit evidence, prepared parts/clamps/protection/checking tools, actual teacher-approved adhesive/process/clean-up. Unconfirmed values remain pending, no guessed times.

Student documents own demonstrated arrangement accurately; clamp pressure is balanced without assumed clamp count, spacing or forces. Protect contact points and maintain alignment.

Actual alignment observation and any approved correction, genuine photo reference, undisturbed storage instruction. Never assert a predetermined clamp duration.

Student page 31: Check surface quality

19. A. Correct. Inspection identifies scratches, glue and uneven areas.

20. B. Correct. Uneven sanding can change the intended form.

21. B. Correct. Follow the authorised sources.

Inspect whole surface to locate scratch and likely cause; unnecessary removal can lose form, round edges or create unevenness. Use approved controlled treatment after teacher advice when needed, remove dust and check again; do not progress on one smooth patch alone.

Student page 32: Check responsible finishing

22. A. Correct. Contamination can create uneven or dirty results.

23. C. Correct. Follow the authorised waste procedure.

24. A. Correct. Durable, maintained work extends material life.

Prevents contamination and avoidable discarded material/rework; avoids over-application, unevenness and wasted finish; longer useful life reduces unnecessary replacement. All quantities, products and waste remain teacher-controlled.

TEACHER GUIDE / BREADBOARD

Pages 33–34: answers and evidence

Student page 33: Explain your decisions

Expected content: full inspection, systematic sanding, preserve form, dust control, quality check. I will inspect the complete bread board under good light and feel the surface for scratches, dents, glue residue or uneven areas. I will sand systematically with the grain where appropriate, working across the whole component rather than concentrating on one spot. I will protect the planned edges and curved end shape by avoiding unnecessary rounding or excessive pressure. I will keep the area clean and manage dust using the teacher-approved method. Before finishing, I will check the complete surface again and ask the teacher to confirm that it is ready.

Expected content: authorised sources, clean protected timber, controlled quantity, secure materials, safe waste. Before collecting the approved oil finish, I will make sure the workspace is clean and the sanded Bread Board is protected from dust, moisture and fingerprints. I will follow the teacher demonstration, product label, SDS and local workshop procedure. I will use only the amount required for the demonstrated stage because excess finish can create waste and an uneven result. I will keep containers secure and avoid returning dirty material to clean finish. Oily rags and contaminated waste will be handed over or placed exactly where the teacher directs; I will not use an ordinary bin or invent another disposal method.

Student page 34: Map the surface condition

Own inspection map identifies actual faults and complete systematic coverage including edges. It is a record, not an invented abrasion direction prescription.

Real observed surface faults/areas, teacher-approved sanding/preparation stage, dust control and repeated inspection. Protect form and avoid invented grit sequence.

Whole-surface observation under good light/feel as appropriate, no residue/dust/unexplained fault, preserved planned form, teacher readiness confirmation where required.

TEACHER GUIDE / BREADBOARD

Pages 35–40: answers and evidence

Student page 35: Prepare finish and waste control

Specific actual product/label/SDS and school directions. Where not supplied record teacher to confirm, not an invented product, PPE specification, ventilation rate or disposal technique.

Prepare first; use only demonstrated amount; clean timber, secure clean containers, no dirty material returned without permission; quality and reduced waste connected.

Authentic action/reason/result and real photo reference; includes safe teacher-directed waste handling and no private details.

Student page 39: Check the evaluation

25. B. Correct. It identifies specific observable evidence.

26. B. Correct. These are the approved comparison points.

27. B. Correct. It explains what and how to improve.

Accept a specific criterion and observable evidence, e.g. hypothetical: both end joints meet without a visible gap on the inspected face. Reject invented claims about own work, unsupported perfection or personal preference alone. A limitation or next check can be stated.

Student page 40: Check evidence and care

28. B. Correct. It documents a meaningful authentic stage.

29. B. Correct. It interprets the evidence clearly.

30. B. Correct. This is responsible shared-workshop practice.

Must identify genuine/hypothetical stage, authorised action, why it mattered and an observable result. Example: during glue-up I checked alignment as balanced pressure was added so the verified fit was preserved; the faces remained level at the check. Do not claim photographic proof of invisible events.

TEACHER GUIDE / BREADBOARD

Pages 41–42: answers and evidence

Student page 41: Feedback that changes work

Expected content: roles, feedback given, feedback received, one improvement, check or evidence. I was the maker first and my classmate was the reviewer; we then swapped roles. The specific feedback I gave was that one end profile looked less even when viewed from the front, so it should be compared with the plan before any more material was removed. The feedback I received was that a small area near one joint still showed glue under strong light. I used roles rather than my classmate's full name to protect privacy. The source model is incomplete: also require a change made or planned because of feedback and a specific subsequent check/evidence of its effect. Accept a planned change identified as future; do not invent completed action.

Authentic own-work feature discussed in teacher-approved peer activity; label links the received/given feedback to an observable check or planned check.

Student page 42: Evaluate your result

Expected content: plan or brief, specific strength, specific fault, likely cause, realistic improvement. Compared with the project plan and brief, one strength is the overall alignment of the curved end pieces. The joints meet closely and the board sits without visible movement, which suggests the dry-fit and clamping checks were effective. One area for improvement is a small uneven patch in the finish near one edge. This may have resulted from inconsistent surface preparation or contamination before the finish was applied. Next time I would inspect the surface under stronger light, clean the area as directed and complete a final teacher check before finishing.

Own feature accurately represented and tied to criterion, observation and limitation; no fabricated quality result or dimension.

TEACHER GUIDE / BREADBOARD

Pages 43–46: answers and evidence

Student page 43: Choose authentic evidence

Expected content: two meaningful stages, action and reason, quality or safety result, honest sequence, privacy. My first evidence item would be a clear photograph of the dry assembly before adhesive. Its caption would explain that I checked the curved end pieces for orientation, close joint fit and alignment with the plan. My second item would show the completed surface under good light, with a caption explaining the final quality check. I would place the images in their genuine project order and keep the original dates where available. I would crop out faces, student names, class lists and unrelated classmates' work so the folio shows only my project evidence.

Two genuinely different meaningful stages; chronological order, action/reason/result and privacy considered. Sketches plan the evidence and do not replace required photographs.

Student page 44: A final evidence panel

Authentic own-board panel, accurate observation and criterion link. A limitation can be evidence uncertainty; do not force a fabricated defect or claim performance from appearance.

Specific strength and limitation compared with plan/brief/teacher criteria; likely cause labelled as inference, realistic next action and check. Photographic evidence has honest date/sequence.

Genuine photo/check/record tied to claim; teacher-approved observation, no invented load or safety test.

Student page 45: Build the evidence index

All twelve source checkpoints point to real, relevant own-work evidence or honest missing/pending status. Inspect sequence, captions, privacy and correspondence. No borrowed illustrative image counted as student proof.

Student page 46: Notice, act, check again

Specific real problem/uncertainty with consequence tied to approved form, fit, surface or safety; evidence separated from inferred cause.

Clear authentic before state/observation. No invented exact defect or project measurement.

Approved correction responds to identified cause, recheck demonstrates effect or remaining uncertainty; no unauthorised machining instruction.

Realistic future action connected to cause/evidence such as earlier datum, fit or surface check; not vague extra care alone.

TEACHER GUIDE / BREADBOARD

Pages 47–50: answers and evidence

Student page 47: Learn from the demonstration

Actual named module/deck/video/demonstration and relevant focus, not generic watched video statement. Practical authority remains teacher.

Meaningful accurate relationship from actual resource, no copied invented machine setup. Possible datum, biscuit matching, clamp/alignment or finish-control relationship.

Specific conceptual learning and relation to own task; no claim that viewing authorises practical operation.

Actual meaningful remaining question or evidence-based confirmation; no invented settings or assumptions.

Student page 48: Keep a useful practical record

Dated genuine stage/action/reason/result, with checks and changes where relevant; preserve actual sequence and omit personal details. Entries need not be lengthy if meaningful.

Concrete source/teacher-controlled next step and prerequisite check; unresolved issue stated accurately.

Student page 49: Care and handover record

Actual correct storage, reported faults without unauthorised repairs, teacher-approved cleaning/waste, safe project/record location. Note unresolved items honestly.

Teacher/product-specific care; no generic washing/oil interval or food-safety claim. Confirmation needed if actual finish/care unknown.

Action-effect connection such as approved cleaning/drying or damage protection supports durability/reduced unnecessary replacement; no guarantee.

Student page 50: Your next piece of work

Three specific actual examples with evidence: suitable hazard/stop response; datum/alignment/surface check; responsible selection/contamination/waste/durability. Not unsupported self-ratings.

Reasoned judgement supported by an actual recorded observation and impact on safety/accuracy/quality/resources; acceptable alternatives depend on own evidence.

Memorable accurate check tied to a real lesson, such as same datum or complete dry fit. No new construction dimensions or unsafe shortcuts.